Machine Learning Based Prediction of Peak Load in EV Charging Stations

Applied Research Project submitted in partial fulfilment of the requirements for the degree Of

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Submitted By

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Declaration

I declare that this Applied Research Project that I have submitted to Dublin Business School for the award of M.Sc. Business Analytics is the result of my own investigations, except where otherwise stated, where it is clearly acknowledged by references. Furthermore, this work has not been submitted for any other degree.

Signed: Rahul Prakash Pagar

Date: 26/08/2025

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Abstract

Electricity distribution networks face many difficulties as a result of the growing popularity of electric vehicles (EVs), especially when it comes to infrastructure planning and station level spinning reserve allocation. The temporal and spatial variability of charging demand is not captured by conventional methods that rely on averages or static growth trends. This study creates a machine learning-based framework for predicting monthly peak electricity loads at EV charging stations in order to close this gap. Weather variables obtained from the Open-Meteo API, temporal and seasonal features, and historical charge data are all included into the system. Several models are trained, verified, and evaluated over various time divisions, including Support Vector Regression, Random Forest, XGBoost, and Neural Networks. MAE, RMSE, and R^2 are used to assess model performance. With temporal variables showing up as stronger predictors than weather conditions, the study shows that Support Vector Regression provides the most dependable generalisation. The results give grid operators useful information to direct infrastructure planning and reserve allocation.

Keywords: Electric Vehicles, Peak Load Forecasting, Machine Learning, Support Vector Regression, Grid Planning, Load forecasting, Ensemble learning

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Chapter 1: Introduction

1.1 Background

The increase in the electric cars (EVs) has come with opportunities and challenges to the modern energy systems. As the world faces global commitments of decarbonisation and minimising greenhouse gas emissions, the EVs have gained momentum with government incentives, reduced cost of batteries, and awareness of people on sustainable transport options (International Energy Agency, 2022). Although EVs are a part of cleaner mobility, their large-scale volume can cause issues with grid stability and peak demand as well as optimal allocation of energy (Zhang et al., 2023).

Charging patterns of EV users are deeply unpredictable and greatly clustered, causing sudden surges in demand at some points in time or during particular seasons. These spikes may cause voltage fluctuations, overloading of transformers and subsequent additional costs of utility companies. To counter these problems, precise prediction of EV charging demand has become more essential to grid operators, city designers, and policymakers.

Conventional statistical techniques have been used to do load forecasting but their inability to handle non-linearities and non-trivial dependencies limits their applicability. New advances in machine learning (ML) and deep learning (DL) have provided hopeful solutions, and models have been able to base their learning on a large pool of data and identify patterns that depend on time, space, and even weather. Machine learning methods have shown to be more accurate in forecasting peak loads, including support vector regression, random forests, gradient boosting, and neural networks (Kong et al., 2017). The importance of data-driven solutions to handle the changing relationship between the EV adoption and electricity demand can be explained by these technological advancements.

1.2 Business Problem

The rapid growth in the number of electric vehicles (EVs) is transforming the electricity demand pattern, causing an unprecedented challenge to the grid network operators. The traditional forecasting methods are not very effective to reflect the non-linear and spatially discontinuous nature of EV charging demand and thus bring risks of over-provision or under-provision of serves or even excessive infrastructure build-up (Aduama et al., 2023). Although their activation can be done within a few seconds, their long-term provisioning and optimum location demand reasonably accurate forecasting of the developing demand patterns (Kong et al., 2017). Utilities are in need of predictive solutions that not only promote operational stability but also lead to allocation of resources, strategic planning, and budgeting to upgrade the infrastructure. This study fills this business-critical gap by leveraging advanced machine learning models to predict peak load demand at EV charging stations to support short-term resilience as well as long-term investment plans in grid maintenance.

1.3 Research Aim

The research aims to predict peak load in electric vehicle charging stations with the help of machine learning models. Through evaluation of past charging data, centre factors of demand variability are determined to help grid operators in long-term planning and energy management efficiency actions.

1.4 Research Questions

Based on our findings, this research will answer the following research questions

1. What is the predicted peak electricity demand (kWh) at EV charging stations when considering historical consumption and weather-related factors?

2. Which input variables such as temperature, humidity, precipitation, wind speed, or seasonal effects contribute most to fluctuations in peak load?
3. Among SVR, Random Forest, XGBoost, and Neural Networks, which modelling approach delivers the most accurate predictions of peak load?
4. To what extent does predictive performance improve when models are optimized with hyperparameter tuning?
5. What are the levels of R^2, RMSE, and MAE obtained by each model in training, validation, and test datasets during the prediction of peak load?
6. What useful information can a grid operator derive concerning infrastructure planning and the assignment of reserves based upon the forecasted peak loads?

1.5 Research Objectives

1. To forecast the highest amount of electricity consumption (peak load) at the EV charging points based on the past charging and weather data.
2. Determine and design the crucial characteristics (e.g., weather, temporal, station characteristics) that impact electricity energy demand, evaluate the interactions among these characteristics.
3. To develop and test several machine learning models, such as SVR, Random Forest, XGBoost, and Neural Networks, on the problem of peak load forecasting and to choose the most suitable method to the data.
4. To train and optimise models systematically by tuning hyperparameters to enhance predictive performance and reliability.
5. To compare the performance of the models in terms of R^2, RMSE, and MAE on training, validation, and test datasets.
6. To derive both theoretical and practical conclusions from the peak load prediction findings, contributing to academic research on energy forecasting while providing

actionable insights to grid operators, policymakers, and energy planners for strategic decision-making.

1.6 Significance of the Research

The importance of this research is in contribute to the emerging environment of electric vehicle (EV) load forecasting, which is essential to sustainable and resilient energy systems. This can be used to make informed peak load predictions which can then be used to guide strategy by the grids in allocating spinning reservoirs, infrastructure and integrating renewable energy into energy grids. Prerequisite literature illustrates that machine and/or deep learning techniques outperform conventional statistical models in terms of predicting non-linear consumption patterns that are driven by weather patterns, time, and behaviour factors (Zhou et al., 2019; Zhang et al., 2023). The discrepancies still exist when it comes to matching model results with practical demands of utilities, such as in localised demand fluctuations at charging points (Zhang et al., 2023).

This paper uses sophisticated algorithms, like Random Forests, SVR, XGBoost and neural networks, on EV charging datasets, and therefore provides utilities with potentially useful tools to predict demand variations more accurately. In addition, it reinforces the academic discussion, through explainability and comparison to baseline methods and without resulting in any compromises of predictive accuracy. This two-fold emphasis promotes both technical and policy-making decisions, which explains the interest of the study to the clean energy transition.

1.7 Chapter Overview

Chapter I :
Introduction

Presents background, business problem, research aim, objectives and significance of predicting EV charging peak load

Chapter II :
Literature Review

Reviews prior work on load forecasting, EV charging demand, and machine learning approaches, identifying research gaps

Chapter III :
Methodology

Describes dataset, preprocessing, feature engineering, modelling with SVR, Random Forest, XGBoost, and Neural Networks, and evaluation metrics

Chapter IV :
Results

Presents and compares model performance on training, validation, and test sets to identify the most effective approach

Chapter V :
Discussion

Interprets results, connects findings with literature, highlights implications for grid operators, and discusses limitations

Chapter VI :
Conclusion and Future work

Summarizes contributions, outlines limitations, and suggests directions for future research.

Figure 1 : Chapter Overview

Chapter 2 Literature Survey

Electricity infrastructures now face additional difficulties as a result of the shift to electrified transportation, especially when it comes to running both public and private EV charging stations. Grid operators must prepare for significant increases in electricity consumption due to the fast-increasing global adoption rates of EVs (International Energy Agency, 2022). EV charging demand is highly clustered and non-linear, usually happening during specified times of day or under specific environmental circumstances, in contrast to traditional home or industrial loads (Zhang et al., 2023). Peak loads brought on by such clustered demand usually put stress on transformers, raise operating expenses, and necessitate the addition of spinning reserves in order to preserve dependability (Islam et al., 2023).

Therefore, predicting the demand for EV charging is a crucial task for legislators and distribution system operators. But a large percentage of earlier research has focused on short-term timeframes, usually hourly or daily load forecasting (Sun et al., 2016). This helps with operational scheduling, but it doesn't immediately help with long-term infrastructure planning. For grid reinforcement and investment decisions, monthly peak load forecast is more important. Additionally, a large portion of the literature has concentrated on deep learning (DL) techniques like convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, which, although accurate, have interpretability and computational cost issues (Aduama et al., 2023).

In order to overcome these constraints, this study uses a variety of machine learning (ML) techniques, including feed-forward neural networks (NN), random forest (RF), XGBoost, and support vector regression (SVR), to anticipate monthly peak demands at the station level. Because of their complimentary qualities in accuracy, efficiency, and interpretability, these models were chosen. ML models like RF and XGBoost, in contrast to black-box deep learning

techniques, offer insights on feature relevance, allowing decision-makers to comprehend the factors influencing demand variability (Wu et al., 2017). This study closes a major gap in the forecasting literature by concentrating on station-level data enhanced with weather and temporal variables. It also offers useful information for grid operators who are organising reserve allocation and infrastructure improvements.

Early research on forecasting power demand mostly focused on statistical models like Kalman filtering, autoregressive moving average (ARMA), and autoregressive integrated moving average (ARIMA) (Zhang et al., 2023). These models were popular in early energy applications because they were clear, easy to understand, and straightforward (Wu et al., 2017). However, their effectiveness in EV applications, where demand patterns are changed by diverse user behaviours and environmental factors, was restricted by their dependence on stationary time series assumptions and incapacity to capture complicated non-linear dynamics. When used on large-scale EV charging networks, simulation-based techniques like Monte Carlo approaches proven to be computationally costly (Zhang et al., 2023).

Research started using machine learning algorithms to identify non-linear tendencies as EV usage increased. One of the first machine learning techniques used for EV load prediction was support vector regression (SVR), which performed well on small-to-medium datasets (Sun et al., 2016). By successfully simulating the intricate interactions between charging demand and meteorological variables, SVR was demonstrated to perform better than ARIMA in short-term EV demand forecasting (Gao et al., 2019). But as datasets get bigger, SVR's computing load increases, restricting its scalability.

Boosting methods such as XGBoost emerged as state-of-the-art approaches in energy forecasting. (Zhang et al., 2023) demonstrated the superior performance of XGBoost compared to SVR and RF across a range of regression tasks (Zhu et al., 2019) applied XGBoost to short-

term EV load forecasting and found it delivered higher accuracy while maintaining robustness against overfitting. Nonetheless, XGBoost, like RF, has primarily been applied to short-term horizons in the literature.

Deep learning techniques also became more popular at the same time. Early EV load prediction research used artificial neural networks (ANNs) (Zhou et al., 2020). More sophisticated LSTM networks, which are excellent at capturing temporal dependencies, came next (Islam et al., 2023, Aduama et al.,2023) showed that LSTM with multi-feature data fusion produced prediction errors as low as 3.29%, while Zhou et al. (2019) showed that LSTM models considerably beat SVR in daily EV load prediction. Short-term accuracy was further enhanced using hybrid models that combined CNNs and LSTMs (Zhou et al., 2020). Despite their effectiveness, DL models are not often used in practical situations since they are frequently computationally demanding, necessitate big datasets.

Therefore, despite much research on EV load forecasting, most is focused on short-term horizons and deep learning methods, and little on monthly station-level peak forecasting with interpretable ML. This study uses four machine learning models SVR, RF, XGBoost, and NN chosen for their shown usefulness in load forecasting and complementing capabilities in light of the shortcomings of previous research. Forecasting energy and EV demand has made extensive use of support vector machines (SVR), an extension of SVM for regression problems. Its capacity to use kernel functions to simulate intricate non-linear interactions is its main strength.

SVR is better than traditional statistical models, according to several research. SVR outperformed ARIMA-based models in EV load forecasting, for example, with an R^2 (R square) of 0.87 and RMSE of 1.92 kWh (Sun et al., 2016). According to other studies, SVR predicted daily EV charging demand with a 12% lower mean absolute percentage error (MAPE) than

multiple linear regression. These findings demonstrate how well SVR captures weather-driven variations in charging behaviour.

Random forest is a collection of decision trees which are well-known to be robust and, at the same time, understandable (Huang et al., 2020). Unlike SVR, RF gives feature importance values and thus practitioners may determine the most significant variables in making forecasts.

Random Forest has proven to be very powerful in EV load forecasting. Experiments with station-level data showed that RF performed better than the linear regression and ANN baselines with an R^2 (R square) of 0.91 (Huang et al., 2020). Other research established that RF was capable of reducing MAE 15 percent compared to multiple regression to forecast charging demand in Shanghai (Zhang et al., 2023). In these results, RF is found as highly efficient with heterogeneous data that contain numerous interacting features like seasons, weather and location.

The effectiveness and regularisation qualities of XGBoost make it one of the most widely used algorithms for regression and classification (Zhang et al., 2023). In contrast to RF, which fits trees independently, XGBoost fits trees sequentially, correcting the residuals of the preceding model with each tree.
Empirical research has linked XGBoost to improved load forecasting efficiency. Specifically, in short-term EV charging load forecasting, XGBoost performed better than SVR and RF, with RMSE improvements of 8% and 5%, respectively (Zhu et al., 2019). Another study that employed XGBoost to anticipate regional loads found that it could learn complicated dependencies, as seen by its R^2 (R square) value of approximately 0.93. One of XGBoost main advantages is its capacity to avoid overfitting by using regularisation parameters (λ, γ), which makes it extremely dependable for datasets that vary between stations and months. XGBoost is a good option for monthly peak demand forecasting at the charging station level since it

strikes a balance between interpretability (via feature importance) and predictive power for this study.

Neural networks are well known to include complex non-linear correlations in data thereby employing them in the forecasting of loads (Chang et al., 2019). In contrast to other deep learning models, such as long short-term memory (LSTM) or convolutional neural networks (CNN), the study utilises a feed-forward neural network (NN) with only one hidden layer, which reflects the approaches taken by other energy forecasting studies to work with the datasets of reasonable size and focus on the interpretability.

The results reported in the previous studies show that even that simple neural nets often produce a competitive performance. As another example, shallow NNs have an R^2 (R square) of 0.89 in forecasting short-term EV loads, and deeper LSTMs further enhanced accuracy (Islam et al., 2023). Comparatively, in the area of hybrid energy systems, NNs used in forecasting had a lower MAPE value of 3.6% than the traditional types of regression processes (Chang et al., 2019). These results demonstrate the conclusion that a feed-forward NN is capable of modelling the non-linear relationship between the EV charging demand and dose, especially when using weather and temporal characteristics.

The use of a feed-forward NN in this study has two purposes; first, it can act as a benchmark of the performance of classical machine learning algorithms (SVR, RF, XGBoost) against a non-linear learner; and second, will understand whether a relatively shallow neural structure can describe station-level monthly peak loads trends without the need to compute resources and data volume as deep learning models would do.

There are still a number of significant gaps in the industry despite advancements: Insufficient emphasis on forecasting medium-to-long-term peak loads: Using deep learning or statistical models, the majority of studies focus on short-term horizons (hourly/daily) (Zhu et

al., 2019; Chang et al., 2019). These methods are helpful for operational control, but they are not sufficient for monthly peak prediction, which is crucial for grid stability and infrastructure development (Sun et al., 2016).

It has been found that much of the literature has concentrated on CNNs, LSTMs, and GRUs to predict EV charging loads (Zhu et al., 2019). Their complexities in terms of computations and data to be handled as well as their black-box problems hinder their implementation in practical applications (Wu et al., 20). This gap informs the usage of more interpretable and reliable ML techniques such as SVR, RF, and XGBoost.

The new studies take advantage of interpretability of models Although RF and XGBoost provide feature importance ranking, much of the literature does not use the ranking to aid decision-making (Huang et al., 2020). This constrains actionable information by grid operators. Most of the studies consider the aggregate regional or national load (Huang et al., 2020). Forecasting at station level, which will be essential when spinning reserve is allocated and at the time of infrastructure upgrading, is limited. This paper fills this knowledge gap by directly utilizing data at station level. Despite the proven effect of weather on charging demand (Islam et al., 2023; Xiong et al., 2015), several researchers consider it as a supporting factor, or even overlook such influence in the monthly peak forecasting process. This study enhances the fusion of weather and time characteristics at station granularity. Literature shows that much progress has been made in EV load forecasting yet there are critical gaps with regard to forecasting range, model interpretability, spatial resolution as well as the importance of weather. Applying the techniques of SVR, RF, XGBoost and NN to the station-level monthly peak prediction, this work has able to bring novel information that is directly applicable to long-term grid planning and reserve assignment.

Chapter 3 Methodology

The approach followed in this dissertation offers a framework to predict monthly high demand of electricity at a station using machine learning (ML) models. The research design takes an **abductive**, quantitative, and comparative perspective, which makes it possible to combine both theoretical knowledge about the problem in the literature and empirical evidence in the dataset to develop findings that would be robust and have a practical focus.

The practicality of the study can be seen in its real-world and pressing energy management problem: predicting peak demand in EV charging stations to ensure that infrastructure planning and related spinning reserve allocation meet the requirements of EV charging. By meeting this challenge, the methodology will not only offer input to academic debate but also can help grid operators, policymakers, and urban energy planners meet the challenge of making the transition to electric mobility a success (Islam et al., 2023).

The research is quantitative because it is based on well-structured numerical data. The central data set consists of station-level transactions of EV charging, such as the energy (kWh) supplied, session duration, plug type, and port type, as well as session start and stop timestamps. These records have been supplemented with weather data, such as the temperature, precipitation, humidity, wind speed and cloud cover which have been received via an external API and therefore offer exogenous variables that have been known to affect EV charging behaviour (Xiong et al., 2015).

The design is both comparative and predictive. It is a comparative as it compares the functionality of five different types of models: Linear Regression (LR), Support Vector Regression (SVR), Random Forest (RF), XGBoost, and feed-forward Neural Networks (NN). The particular models were chosen due to their complementary advantages of LR: being a benchmark model that is easy to interpret, and of SVR: capturing of non-linear relationships

between variables of RF: strong robustness and effortless analysis of feature importance of XGBoost: its efficiency and regularised boosting approach (Zhang et al., 2023), and of NN: its ability to capture non-linear feature interactions (Mahmoud et al., It is also predictive in the sense that the ultimate reference is to predict monthly peak load values with results that are directly applicable in the reinforcing of infrastructure and strategic reserves planning.

By adopting such an abductive comparative research approach, this study aims to fill some of the identified gaps in the current literature, e.g., the lack of focus on medium-to-long-term peak demand forecasting, the lack of station-specific instead of aggregate modelling, and the need to provide explainability and feature importance analysis applicable in real, practical grid applications (Huang et al., 2020). The methodology will therefore aim at coming up with forecasts that are not only accurate but also understandable and, in a position, to be directly implemented in real-life decision-making.

Figure 2 provides a pictorial overview of the methodology design connecting the steps data collection, preprocessing, feature engineering, model training, evaluation, and interpretation into a single coherent pathway.

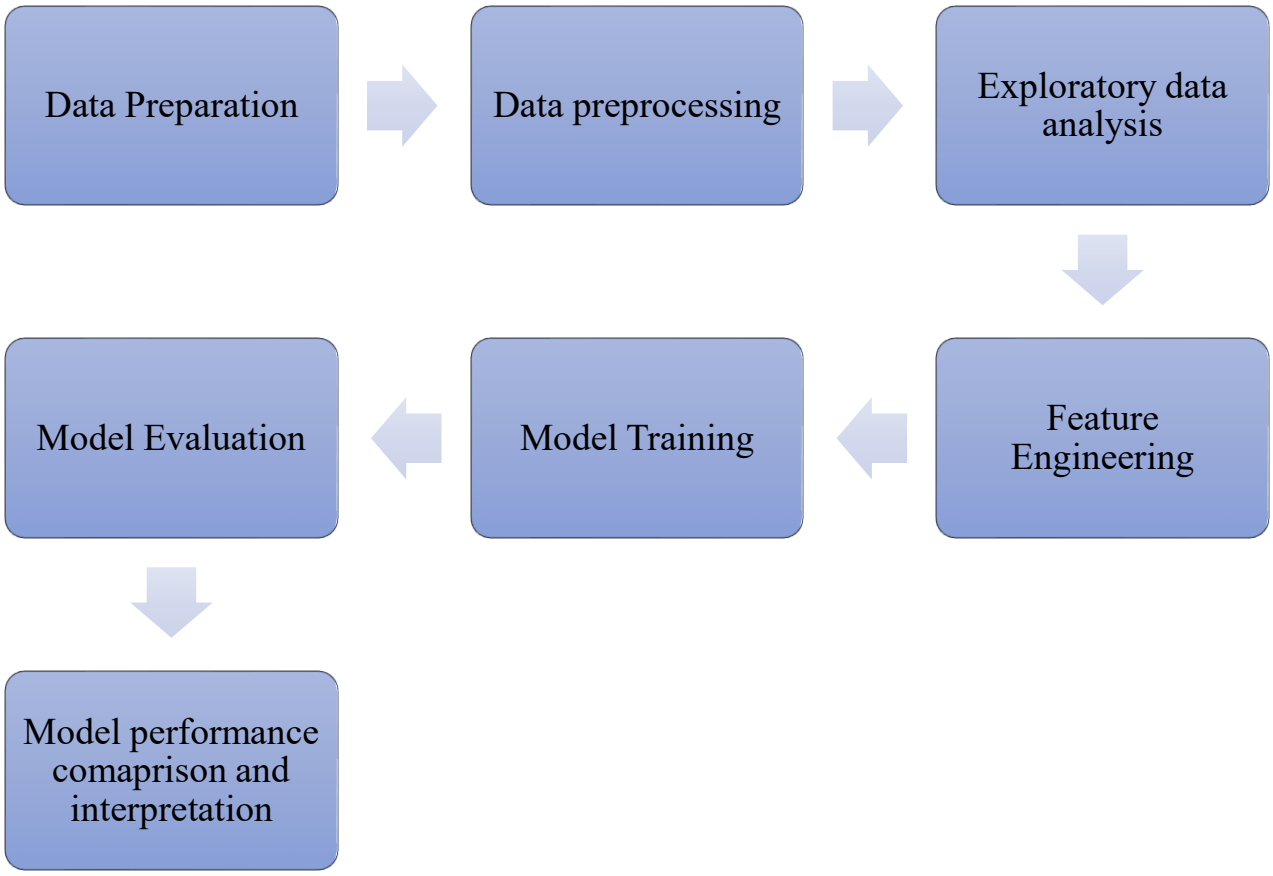

Figure 2 : Methodology for peak load prediction

3.1 Dataset source and Dataset Description

1. Charging station usage dataset:

Data used in the study was downloaded on Kaggle due to EV Charging Station Usage of California City repository. It is a set of ten years of historical charging data at the public charging stations in Palo Alto, California, being one of the richest openly available data sets of EV chargers. The increased temporal scope is useful in this study, owing to its ability to examine the trend of the demand over the long-term period, seasonal pattern, and peak load patterns at the station base.

There is one row of the dataset dedicated to the individual charging session and contains information about the charging process, equipment, and the location. To give a broad direction, the characteristics of the dataset can be summarized into the following categories:

Station and Organisation Information: It contains information like Station Name, Organisation Name, EVSE ID (equipment identifier), Port Number, Port Type (e.g., Level 1, Level 2) and Plug Type (e.g. CCS, CHAdeMO, Tesla). These attributes describe the charging infrastructure taken and identify the alternative structures based on hardware.

Timing and Duration: Examples of attributes in this group are Start Date, End Date, and Transaction Date (Pacific time), which pertain to the time at which charging exercises are conducted. The dataset also contains Total Duration (HH:MM: SS) containing the total connection time, and Charging Time (HH:MM: SS) reflecting the actual time when energy was transported. These temporal variables form the basis of obtaining seasonal, daily and monthly patterns.

Energy and Transaction Details: This group includes Energy (kWh), the primary dependent variable that measures electricity consumption per session, and transaction level variables, such as Fee, Currency and Ended By, which measure whether the session was ended manually or automatically. These characteristics connect usage habits with technical and economic charging characteristics

Geographical Information: The data set includes Address, City, State/Province, Postal Code, Country, Latitude, and Longitude.

User and Equipment Metadata: Some other fields concern user IDs and technical data.

2. **Weather Data via API Integration**

   The addition of weather data represented a significant dataset expansion. Previous studies show that weather has a significant impact on the demand for EV charging. For instance,

low temperatures raise heating needs and decrease battery efficiency, increasing the need for charging (Xiong et al., 2015). The number and timing of charging episodes can also be impacted by driving behaviour, which is influenced by precipitation and cloud cover (Islam et al., 2023).

The Open-Meteo API was used to acquire historical weather data in order to record these effects. High-resolution, location-specific weather variables for a specified time period can be retrieved using this API. Hourly meteorological data was retrieved for this investigation, together with the latitude and longitude that corresponded to the regions of charging stations. Among the parameters that were requested were:

- Temperature (2m above ground, °C),
- Relative humidity (%),
- Precipitation (mm),
- Wind speed (10m above ground, m/s),
- Wind direction (degrees),
- Cloud cover (%).

3.2 Data cleaning Preprocessing:

To ensure analytical robustness, a methodical cleaning and preparation step was carried out after the EV charging station usage information was combined with outside meteorological data. According to Hong and Fan (2016), preprocessing is an essential part of energy forecasting workflows because low-quality data directly affects the stability and performance of machine learning models.

Handling missing values: In order to maintain the chronological sequence that is essential for temporal aggregation, missing transaction timestamps were filled using corresponding End times.

Robust Date Parsing: To accommodate unusual formats, a tolerant parser was used, and temporal information (Year, Month, Weekday, Is Weekend) were extracted.

3.3 Temporal Aggregation and Monthly Dataset Construction:

In order to prepare the dataset, raw charging transactions were aggregated into daily energy totals for each station. Location and DateOnly were used to categorise charging sessions, and the total energy given in kWh was added up to create a new feature called daily_kwh. In this process, significant demand fluctuations were preserved while session-level volatility was smoothed. The forecasting literature generally acknowledges daily aggregation as a transitional stage that strikes a compromise between stability and granularity.

Daily values were rolled up to the monthly level in order to match the dataset with the medium-to-long-term forecasting study focus. Three main indicators were calculated: peak_load_kwh, which is the largest daily demand in a given month; total_load_kwh, which captures the overall monthly throughput. Peak load was chosen as the primary dependent variable, with average and total loads kept as supporting features, because infrastructure design and spinning reserve allocation are mostly based on worst-case stress scenarios.

Through monthly weather integration, external drivers were included. In order to minimise the dimensionality problems associated with high-frequency data and to ensure temporal alignment with demand, hourly meteorological data (temperature, relative humidity, precipitation, wind speed, wind direction, and cloud cover) were combined into monthly averages. Prior research demonstrates that balancing demand and weather resolution.

Finally, energy and weather panels were merged by *Location* and *Year-Month (YM)*. Additional temporal fields, including *Year* and *Month*, were extracted to facilitate trend and seasonality analysis. The resulting structured dataset (*EV_Data_Final.csv*) contained target variables, weather predictors, temporal markers, and cleaned station identifiers, forming a station-level

monthly panel enriched with environmental and temporal features. This dataset provided the foundation for feature engineering, exploratory analysis, and predictive modelling.

3.4 Exploratory Data Analysis and Outlier treatment:

Visualisation and data analysis were carried out following Temporal Aggregation and Monthly Dataset Construction in order to extract significant and meaningful information from the dataset. The monthly peak electricity demand at EV charging stations is supported by temporal trends, regional disparities, and environmental effects that were discovered during this stage. The monthly trend in peak load averaged for all stations from 2011 to 2021 is shown in Figure 3. According to the data, there was a significant growth in EV usage and a greater reliance on public charging infrastructure between 2011 and 2016. The tendency starts to level out in 2017 and even starts to drop in subsequent years, especially in 2019 and 2020. This could be due to COVID-19 regulations and demand saturation. This emphasises how forecasting models must include temporal variables (year and month) in order to account for both cyclical downturns and long-term growth.

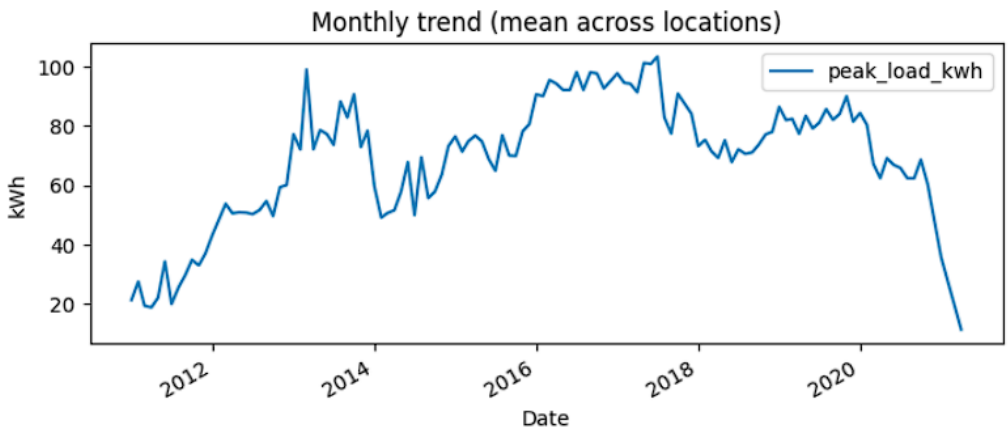

Figure 3 : Long term trend in Peak Load

Figure 4, ranking of the stations on the basis of maximum observed monthly peaks, gives a clear picture regarding spatial variation in demand. Other stations registered much lower peaks constantly, whereas, high-performance ones such as WEBSTER #1, WEBSTER #2 had peaks of more than 190 kWh. This shows that demand of charging differs significantly across locations with some areas acting as key sites. As such, as opposed to relying on predictions to find a balance in aggregate levels, the crucial element is to include station level identifiers to distinguish site-specific behaviour.

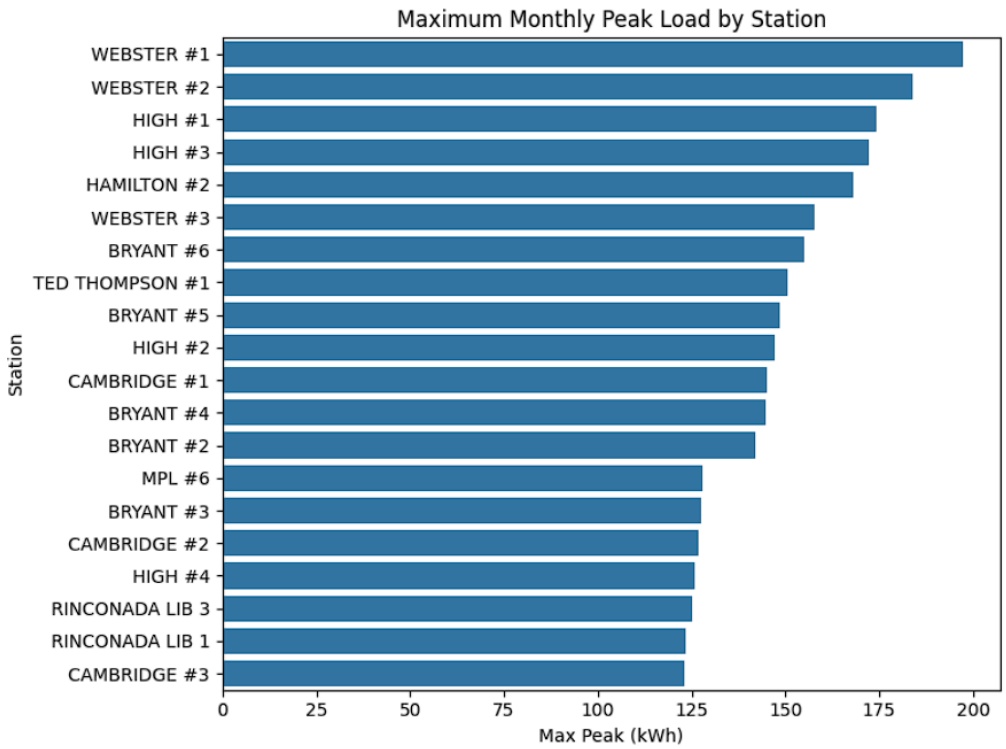

Figure 4: Maximum Peak load by station

Figure 5 shows a Year × Month heatmap that was used to better examine seasonality. According to the data, there are regular cycles of increased demand in the summer and decreased demand in the winter. Growth continued until about 2017, at which point fluctuations became more erratic. Given that seasonal cycles systematically affect peak loads, there is compelling evidence to support the inclusion of seasonal encodings (Spring, Summer, Autumn, and Winter) in addition to weather variables.

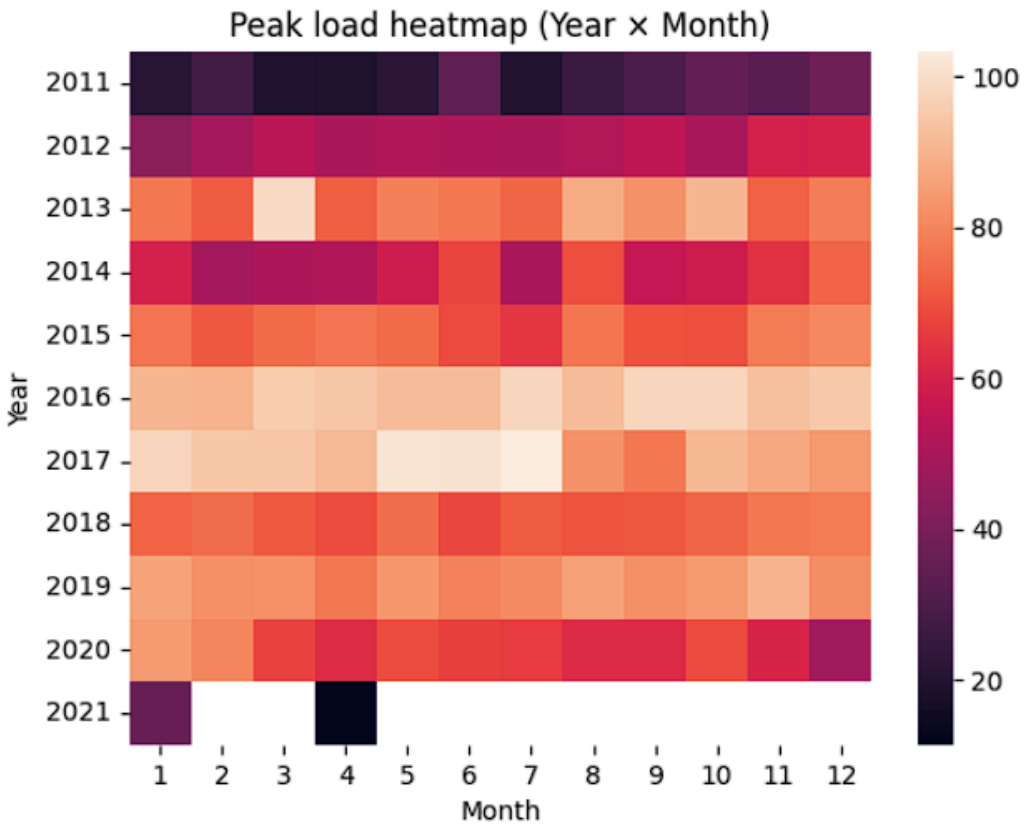

Figure 5 : Seasonal Patterns

During analysis figure 6 illustrates the results of additional investigation into the weather-demand relationship, which revealed non-linear trends between peak load and temperature. Demand rose in the spring and summer when temperatures were moderate, then levelled out or decreased at higher extremes. Autumn and winter, on the other hand, showed less correlation. In order to fully reflect the impact of meteorological factors on load fluctuation, these non-linear dynamic highlights the importance of incorporating polynomial transformations and interaction terms.

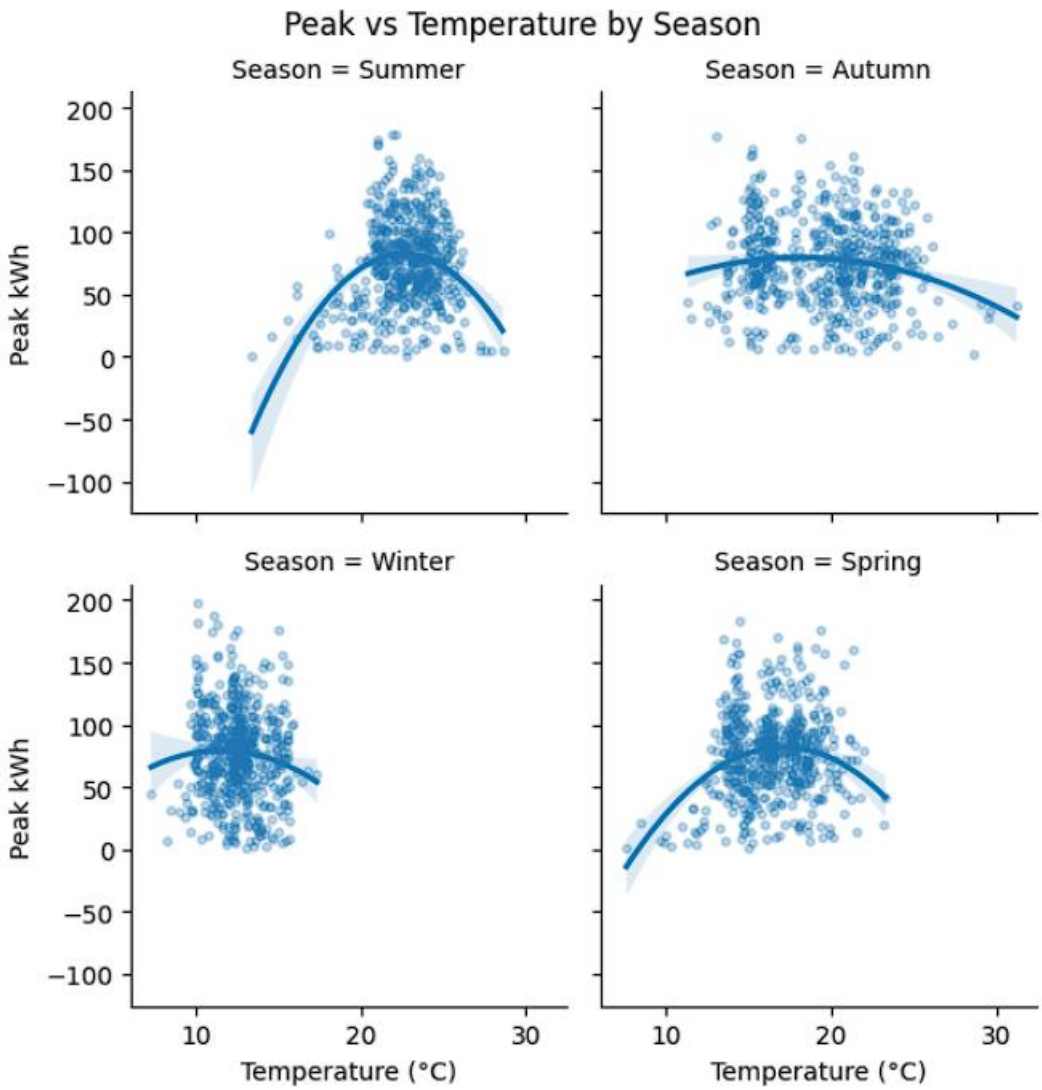

Figure 6 : Weather Demand Relationship

Although the exploratory data analysis identified anomalies in the explanatory variables, it also validated significant temporal, geographical, and climatic trends. Occasionally, weather-based predictors such as wind direction, humidity, and precipitation showed unrealistic numbers, most likely due to recording errors or malfunctioning sensors. Ignoring such outliers could cause bias in estimates and inflate variance, which would skew model training. This was addressed by applying the interquartile range (IQR) approach to continuous explanatory variables, where values outside the range [Q1−1.5×IQR,Q3+1.5×IQR] were trimmed to the appropriate thresholds. Additional physical restrictions were imposed to maintain climatic realism: precipitation must not be negative, wind direction could not be greater than 360°, and

relative humidity and cloud cover could not be greater than 0–100%. Since excessive demand levels indicate real system stress, outlier adjustments were not made to the objective variable, peak load. This preserved critical demand extremes while guaranteeing stable predictors.

3.5 Feature Engineering:

The raw station and weather records were converted into informative and interpretable predictors by applying feature engineering. Based on this evidence and the analysis developed, an efficient combination of temporal indicators, season encodings, variable weather-related factors, and station identifiers was designed to capture the multi-dimensionality of EV-charging requests.

The month was encoded with Winter, Spring, Summer, Fall and one-hot encoding. Fall was chosen as a reference to prevent perfect multicollinearity, and Spring, Summer, and Winter were binary indicators. This representation supports numerical stability and can be used by models like SVR and RF to exploit seasonal effects. Monthly averages of weather variables from the Open-Meteo API were incorporated as covariates.

The Location field was encoded using one-hot indications since modelling is done at station granularity.

Peak load was chosen as the primary target variable, as it most directly informs capacity planning and spinning reserve allocation.

EDA insights and earlier forecasting studies were used to clearly guide the feature engineering approach. Weather, seasonal, and temporal factors guarantee that the models account for both exogenous climate influences and endogenous patterns in charging behaviour. By adding a spatial component, station identifiers allow models to be differentiated across various infrastructure locations. When combined, these engineered properties allow the models to remain interpretable while generalising across a variety of circumstances.

3.6 Model selection and training:

The temporal split technique was used to maintain the data's chronological order, mirroring real-world forecasting where past observations are used to estimate future demand. The dataset was separated into:

Training set (2011–2017): for preliminary model fitting,

Validation set (2018–2019): for hyperparameter tunning and evaluation,

Test set (2020–2021): For final evaluation.

This strategy adheres to best standards in load forecasting research, where generalisability depends on preserving temporal order.

The target variable at prediction was peak load (peak_load_kwh). Peak load directly establishes the system capacity needs as well as spinning reserves.

Linear regression:

Simple Linear Regression (SLR) was included as a baseline model to provide a transparent point of comparison for more advanced approaches. Linear regression models the relationship between a dependent variable and one or more independent variables using a straight-line equation.

$y=\beta 0+\beta 1x1+\beta 2x2+...+\beta nxn+\epsilon$

Linear regression is employed as a baseline benchmark because it models a linear association between predictors and the target and yields directly interpretable coefficients. In the EV peak-load forecasting context, this specification is well suited to capturing broad structural trends (e.g., year-on-year growth) and systematic seasonal variation.

Support Vector regression:

Then second model used Support Vector Regression (SVR), a kernel-based method chosen for its ability to accurately describe non-linear connections in small-to-medium datasets. By finding a hyperplane that minimises error within an ε-insensitive margin and penalising departures outside of this tolerance through a regularised loss, SVR estimates a regression function. The RBF kernel is especially well-suited for non-linear weather–demand interactions, according to an evaluation of many kernels (linear, polynomial, and radial basis function).

Random forest:

The third model in this study was the Random Forest which is an ensemble whose model is made by averaging over datasets of the randomly chosen subsets of both the dataset and the features in an attempt to both reduce the variance and avoid overfitting. Each tree provides a separate prediction and the final result is obtained by averaging, with performance determined by important hyperparameters the number of trees (n_estimators), which balances between the accuracy and calculation time; max_depth, which limits tree complexity; min_samples_split and min in samples_leaf, which regulate the nodes splitting and the size of the leaf; max_features, which is discussed in two schemes sqrt and log2. Random Forest showed advantages in its ability to process high-dimensional inputs -such as one-hot encoded station identifiers- and provide transparent feature importance scores to aid in the explanation of the primary temporal and meteorological drivers of peak load, as well as robustness to outliers by averaging across trees.

XG Boost regression:

The fourth model used**, XGBoost**, is a gradient-boosting framework that iteratively fits decision trees so that each new tree corrects residual errors from the ensemble built so far. Its learning objective combines mean-squared-error loss with explicit regularisation to control model complexity and mitigate overfitting. In this study, the tuning focused on the principal capacity and shrinkage controls n_estimators (boosting rounds), max_depth (tree depth), and

learning_rate (per-tree contribution) together with stochasticity parameters subsample and colsample_bytree to decorrelate trees, and regularisation terms including gamma (minimum loss reduction for splits) and reg_lambda. XGBoost is well suited to tabular EV-load data. it delivers high predictive accuracy, captures non-linear interactions between temporal drivers and weather covariates, and, like Random Forests, provides feature-importance diagnostics that support model interpretability and explanation.

Neural Network:

The fifth model deployed was neural networks as a benchmark extant deep-learning technique against more explainable machine learning models. Similar to prior studies in the field of energy forecasting that use multilayer perceptron's as well as recurrent models the multi-layered neural network can learn complicated interactions between weather, temporal, and station-level predictors. The hyperparameter optimisation was performed on the number of hidden layers and neurons, the learning rate and batch size, activation functions, and regularisation with the dropout technique to prevent overfitting. In this context, the approach has two key merits: the ability to pick up highly non-linear relations among heterogeneous inputs and the ability to scale architecture depth and width to the amounts of available data, thereby serving as a rigorous deep-learning benchmark to the analysis interpretable models in the study.

3.7 Model evaluation and Interpretation:

Assessment of predictive models is also a vital step towards both methodological integrity and practical significance. The proposed evaluation framework was aimed at testing the performance of various models a Simple Linear Regression (SLR), Support Vector Regression (SVR), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Neural Networks (NN) to predict monthly charge peaks at a charging-station level.

Evaluation Metrics

Three measures were used to evaluate the performance:

Coefficient of Determination (R square): It expresses ratio of variance explained by the model in the target. Considerable explanatory power is represented by large R squared values. In peak load forecasting, this can be interpreted as capability of capturing not only the trends over time but also the spatial heterogeneity of peaks.

3.8 Hypothesis

This study formulates the following hypotheses; each framed in null (H_0) and alternative (H_1) form. In keeping with predictive modelling practice, validation is undertaken using model performance metrics (R^2, RMSE, MAE).

In this study, a number of hypotheses were developed each of which was framed with null (H_0) and alternative (H_1) formulations to make it testable. The first (H_1) addresses the level of predictability of the peak electricity demand station wise. The null hypothesis is that historical consumption and weather-related variables are not useful in the prediction of monthly peak loads whereas the alternative hypothesis assumes otherwise. They are validated by finding the values of RMSE, MA E and R^2 of a model on training, validation and test data set. When the predictive accuracy of the model is significantly better than averages, and consistency of range is observed, H_0 will be rejected in favour of H1.

The second hypothesis relates to the contribution of weather and temporal variables. Here, the null assumes that seasonal and climatic indicators such as temperature, precipitation, and humidity are not significant predictors of peak demand, while the alternative suggests that they play a central role. Validation is based on feature importance analysis using Random Forest and XGBoost, as well as correlation tests, to establish whether these factors emerge as dominant contributors.

The third hypothesis is about comparative model performance. It assumes under H_0 that advanced modelling, especially with XGBoost and Neural Networks, does not largely outperform more basic modelling, such as Support Vector Regression, Random Forest, or Linear Regression. The alternative presumes the better outcomes of advanced models, which are tested on the basis of comparing R^2 and RMSE values on a test set.

Lastly, the fourth hypothesis addresses the practical usefulness of forecasts to the grid operators. The null hypothesis denotes that the forecasted highest loads do not provide practical information in planning purposes, whereas the alternative claims the opposite. Validation involves interpreting the outputs in the context of reserve allocation and infrastructure decisions.

Root Mean Squared Error (RMSE): Penalises large deviations between predicted and actual values, making it particularly relevant for peak load prediction where underestimation can threaten system reliability.

Mean Absolute Error (MAE): It offers an easy-to-understand measure of the mean absolute error in kWh.

Using multiple metrics ensured that models were evaluated not only on overall fit but also on their ability to limit extreme errors, which are critical for grid planning (Islam et al., 2023)

The models were trained on the data of 2011-2017 and tuned on 2018-2019 by grid search with cross-validation and again tested on the 2020-2021. This temporal separation guaranteed that models were tested in realistic circumstances of forecasting future demand given a history about demand, that is, elimination of lookahead bias.

Chapter 4 Results

In this chapter the results of the predictive modelling experiments are presented. The hypothesis was to determine the capability of five machine learning models, such as Simple Linear Regression (SLR), Support Vector Regression (SVR), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Neural Networks (NN), to predict monthly peak loads at the level of the EV charging stations. The results are presented both on validation, and test datasets to allow a full picture of how well the model performs. The performance of the training, validation, and test data are reported with three performance metrics: coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE). The results are tabulated by model, and then followed by a comparative summary.

Linear regression results:

Linear regression first implemented. As shown in Table 1, model explained a variance of 85.5% on train data but lesser on validation and test at 60% and 58%, respectively. The errors increased to RMSE 18.90 (validate set) and RMSE 20.81 (test set) and the MAE increased to MAE 15.86 and MAE 16.92.

Table 1: Linear model results

Linear Regression			
Dataset	R square	RMSE	MAE
Train	0.8552	13.96	10.73
Validate	0.6048	18.9	15.86
Test	0.58	20.81	16.92

As shown in Figure7 Linear Regression has failed to capture the sharp variations of peak demand, especially at high loads. Multiple underestimations can be observed when the actual

demand peaked higher than 120 kWh. Figure 5.2 (Prediction Parity) demonstrates its tendency to cluster around the diagonal with a systematic underprediction at higher values.

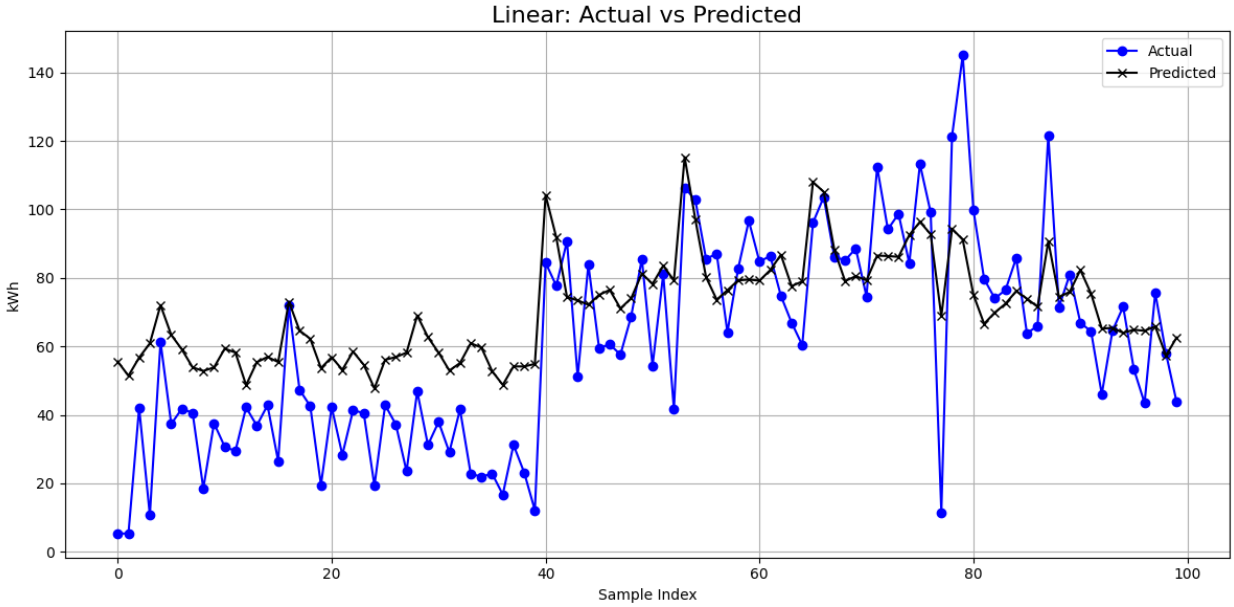

Figure 7 : Linear Regression Actual vs Predicted Peak Load

Figure 8 (Prediction Parity) is clustered close to the diagonal but with systematic underprediction as the response increases.

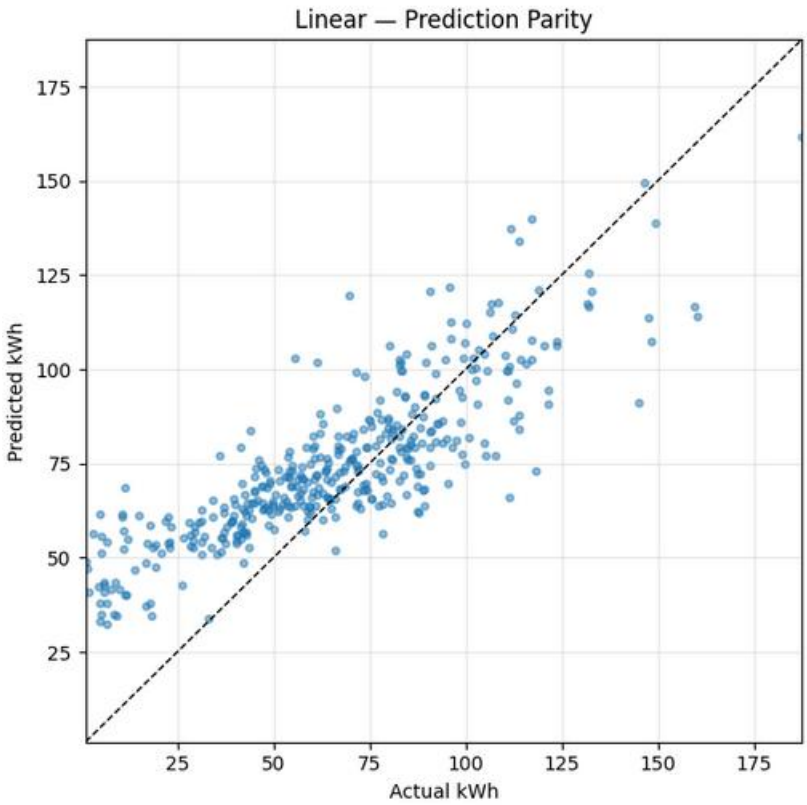

Figure 8 : Linear Model Parity Plot: Actual vs Predicted Peak kWh

SVR results:

As shown in Table 2, SVR compared to Linear Regression improved slightly Training R^2 was 0.8493, then the validation R^2 and test R^2 was 0.6257 and 0.6063 respectively. The RMSE values were 14.24 (train set), 18.39 (validate set), and 20.33 (test set) and the MAE was 10.46, 15.34, and 16.55 respectively.

Table 2: SVR model results

SVR			
Dataset	R square	RMSE	MAE
Train	0.8493	14.24	10.46
Validate	0.6257	18.39	15.34
Test	0.6063	20.3367	16.55

Figure 9 indicates that SVR tracked the medium-sized peaks better than the Linear Regression model, but, nevertheless, failed to capture the high spikes well.

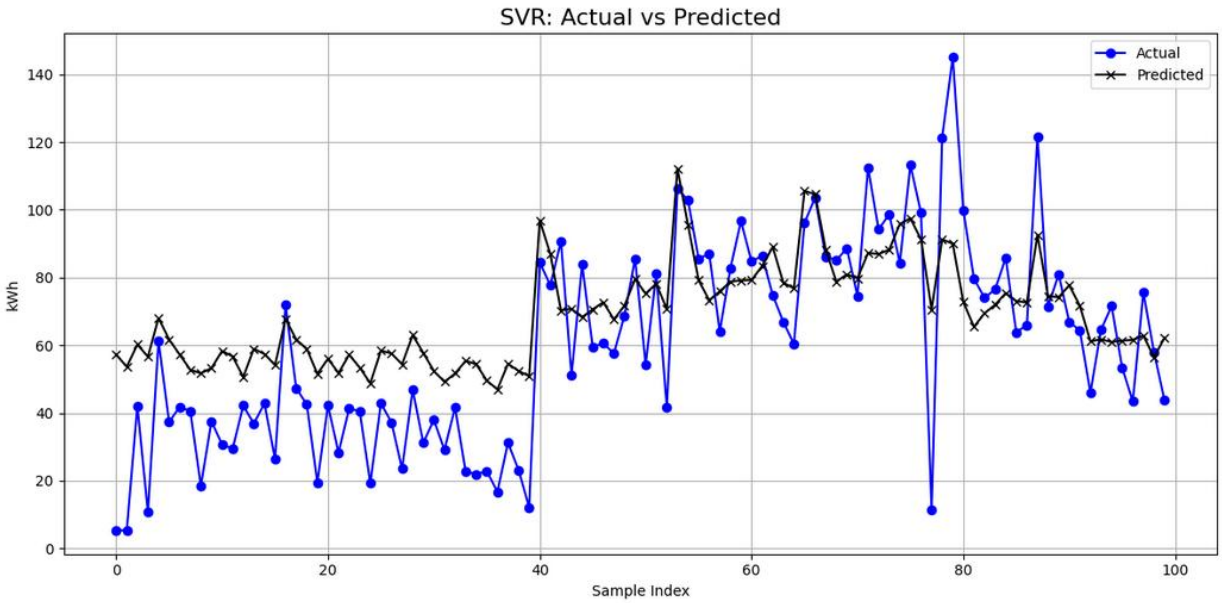

Figure 9: SVR Actual Vs Peak Load

Figure 10 shows a tighter clustering around the diagonal than Linear Regression, suggesting improved accuracy across moderate ranges.

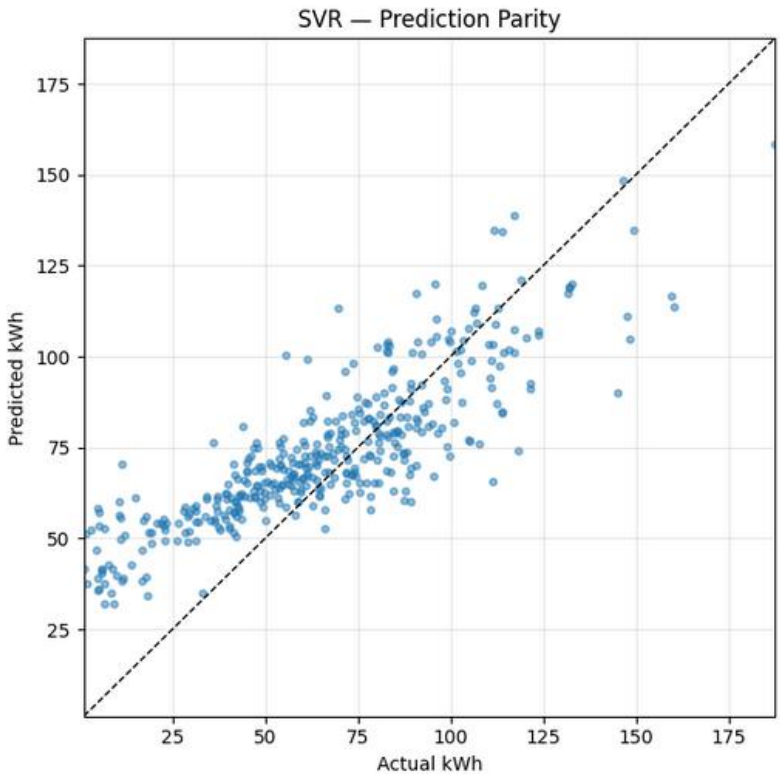

Figure 10 : SVR Model Parity Plot: Actual vs Predicted Peak kWh

Random Forest results:

As shown in Table 3, Random Forest Regression produced strong performance on the training set, achieving $R^2 = 0.8915$ (RMSE = 12.08, MAE = 9.59). Validation R^2 was 0.6427 (RMSE = 17.97, MAE = 14.62), while test R^2 reached 0.5571 (RMSE = 21.57, MAE = 17.64).

Table 3: RF model results

Random Forest			
Dataset	R square	RMSE	MAE
Train	0.8915	12.084	9.59
Validate	0.6427	17.97	14.62
Test	0.5571	21.57	17.64

Figure 11 indicates that the Random Forest model represents the general pattern of time series demand but removes sharp turns, does not find extreme peaks.

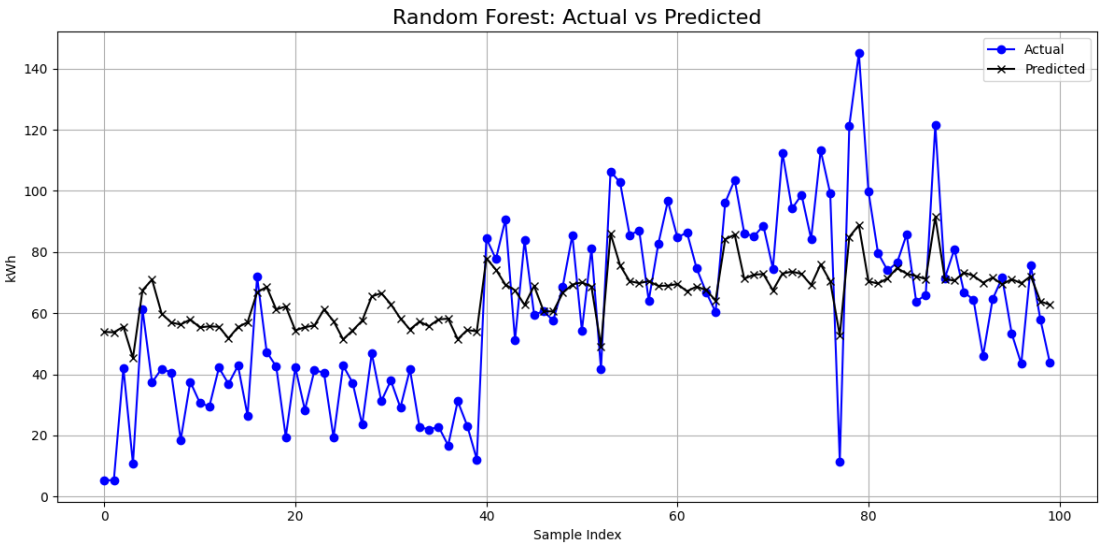

Figure 11: Random Forest Actual Vs Peak Load

Figure 12 confirms a systematic bias towards underprediction at higher loads, with most of the points below the diagonal line.

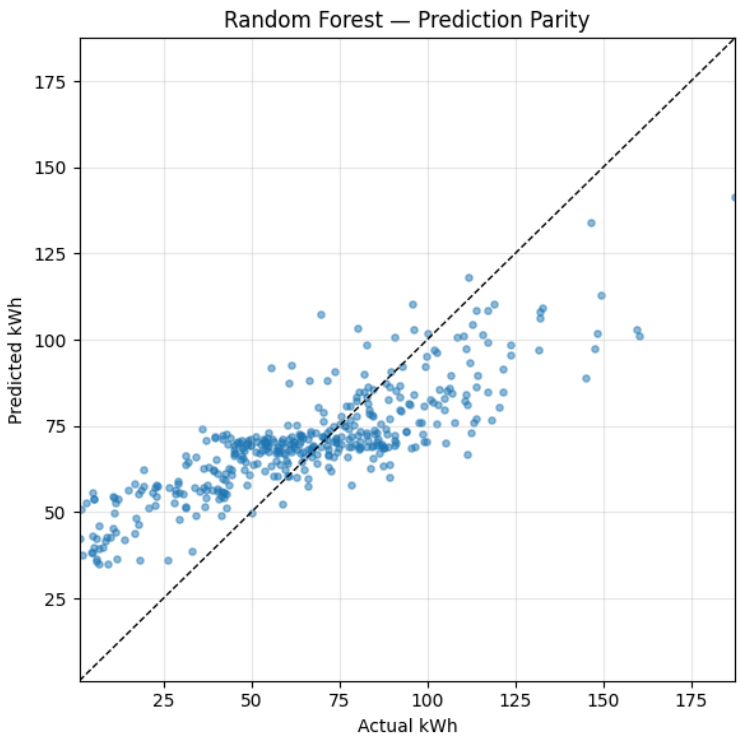

Figure 12 : RF Model Parity Plot - Actual Vs Peak Load

XG Boost results:

As shown in Table 4, XGBoost delivered the strongest validation performance among all models. It achieved an exceptionally high training R^2 of 0.9980 (RMSE = 1.64, MAE = 1.20), indicating near-perfect fitting of training data. Validation R^2 was 0.6900 (RMSE = 16.74, MAE = 13.48), while test R^2 reached 0.6119 (RMSE = 20.19, MAE = 15.60.

Table 4: XG Boost Results

XG Boost			
Dataset	R square	RMSE	MAE
Train	0.998	1.64084	1.1954
Validate	0.69	16.74	13.48
Test	0.6119	20.19	15.6

Figure 13 shows that XGBoost followed demand fluctuations more closely than Random Forest, capturing peaks more accurately.

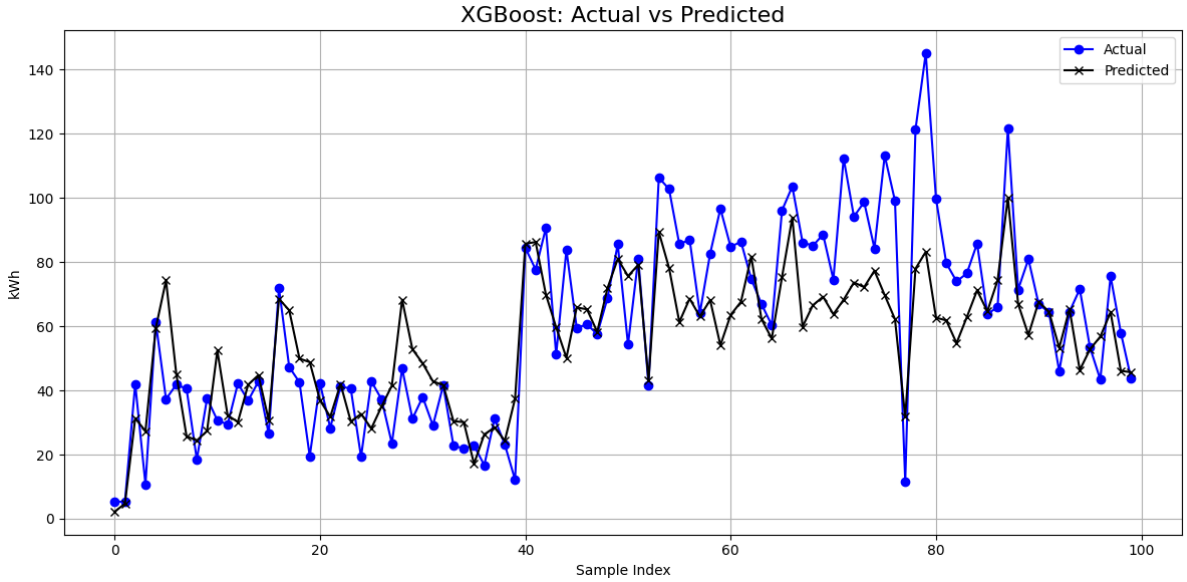

Figure 13: XG Boost Actual Vs Peak Load

Figure 14 shows a closer adherence to the diagonal, reflecting a broader adherence to predictive accuracy.

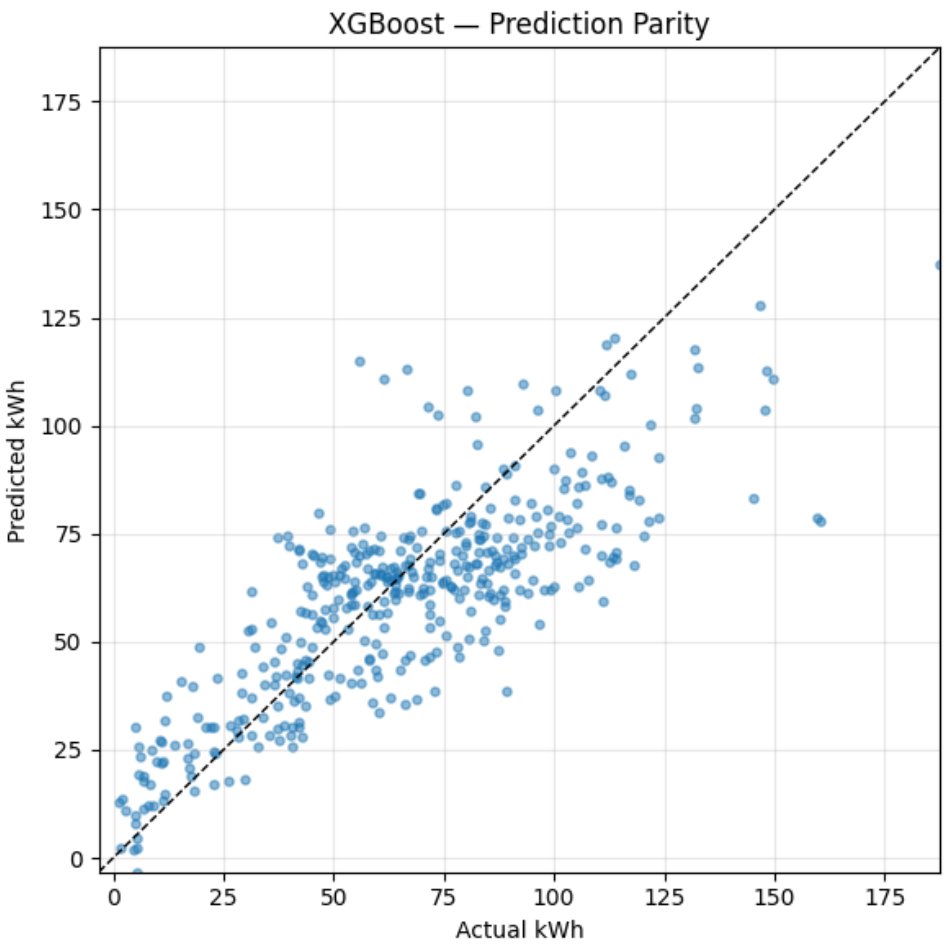

Figure 14: XG Boost Model Parity Plot - Actual Vs Peak Load

NN results:

As shown in Table 5, the Neural Network produced high training accuracy, achieving a training R^2 of 0.8932 (RMSE = 11.98, MAE = 9.21), but struggled to generalise. Validation R^2 was 0.5592 (RMSE = 19.96, MAE = 16.41), while test R^2 reached 0.4898 (RMSE = 20.49, MAE = 18.59).

Table 5: NN Results

Neural Network			
Dataset	R square	RMSE	MAE
Train	0.8932	11.98	9.20
Validate	0.5592	19.96	16.41
Test	0.4898	23.15	18.59

Figure 15 illustrates that the Neural Network predictions were overly smoothed, missing sharp changes in peak demand.

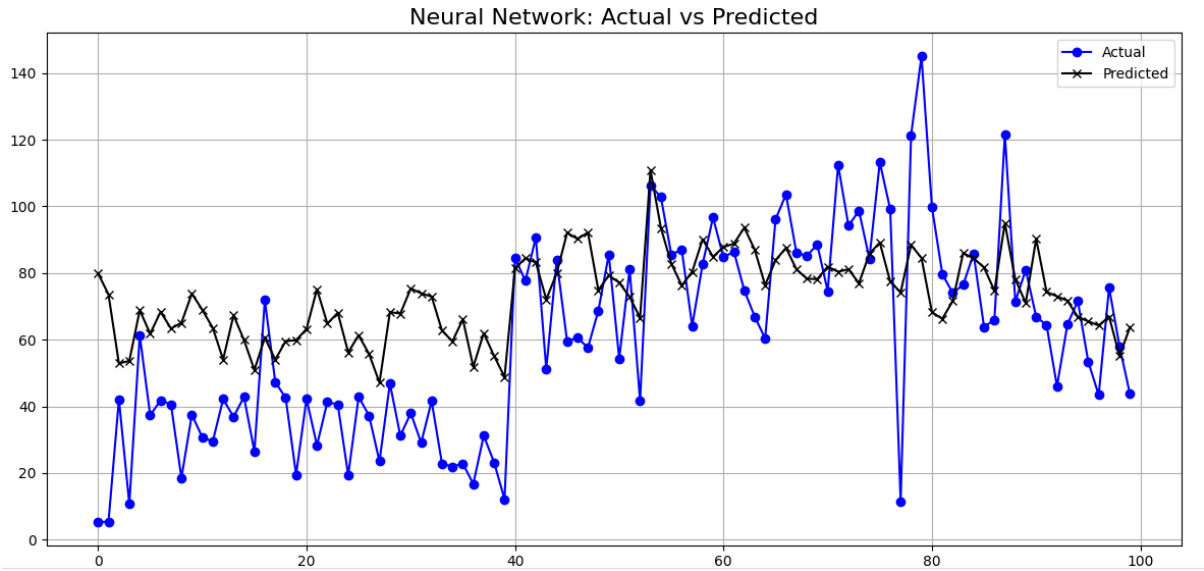

Figure 15: NN Actual Vs Peak Load

Figure 16 shows widespread scatter around the diagonal, especially for higher demand levels.

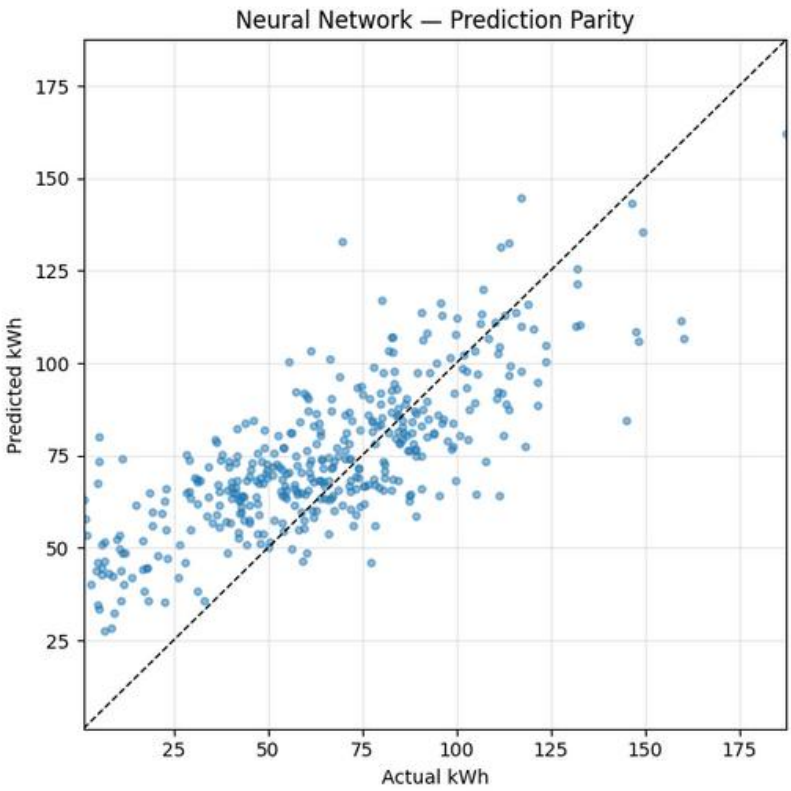

Figure 16 : XG Boost Model Parity Plot - Actual Vs Peak Load

Comparative Model Analysis

As shown in Table 6, the performance of Linear Regression, SVR, Random Forest, XGBoost, and Neural Network is compared across training, validation, and test sets using R^2 and RMSE.

Table 6: Comparative summary

Model	Train R^2	Val R^2	Test R^2	Train RMSE	Val RMSE	Test RMSE
Linear Regression	0.8552	0.6048	0.58	13.96	18.9	20.81
SVR	0.8493	0.6257	0.6063	14.24	18.39	20.3367
Random Forest	0.8915	0.6427	0.5571	12.084	17.97	21.57
XG Boost	0.998	0.69	0.6119	1.64084	16.74	20.19
Neural Network	0.8932	0.5592	0.4898	11.98	19.9646	23.15

As shown in Table 6, the XGBoost method provided the best validation R^2 (0.6900) and therefore the best approximation of generalisation at the time of the model-selection. SVR attained the best performance on the test set in this comparison R^2 (0.6063), followed closely by XGBoost (0.6119). SVR also had the least variation across splits, with a relatively consistent train, validation, and test performance. On the other hand, XGBoost was distinctly overfit, with an almost perfect training fit and a significant drop on validation and test. The Neural Network had the worst model performance because it did not generalise and it had the lowest test R^2. The findings indicate that at the station level, XGBoost and SVR models are the most promising models to forecast monthly peak loads in terms of accuracy and generalisability. Random Forest showed fair but biased performance on extreme values. Linear Regression was a practical starting point yet was incapable of fitting complex relationships. Neural Networks did not perform well because of small size of data used and the restriction to optimise.

Chapter 5 Discussion

5.1 Introduction

This chapter interprets the findings presented in previous chapter and contrasts them with the expectations set out in the methodology. The intention is to critically reflect on the predictive performance of the models, the significance of feature groups, and the practical implications for grid operators. Unlike the results chapter, which presented model outputs, this chapter goes further by comparing what was anticipated with what was observed, situating the findings in the context of existing literature, and analysing their implications for medium-to-long-term forecasting of peak electricity demand at EV charging stations.

5.2 Comparison of Anticipated and Observed Outcomes

This study initially assumed that complex algorithm models like XGBoost and Neural Network will perform better than the simple models like Support Vector Regression (SVR), Random Forest (RF), or Linear Regression. The expectation was based on the high degree of agreement in the previous literature that deep learning and boosting models performed best in energy forecasting due to their capability of modelling complex non-linearities and high-dimensional relationships. Moreover, the combination of weather and time-related factors was likely to display far better predictability outcomes, and temperature, precipitation, and humidity were projected to be the determining factors when it comes to launching the peak demand (Islam et al., 2023; Sun et al., 2016).

These expectations were not met in key aspects by the actual findings. XGBoost and Neural Networks performed notably well during training but poorly during testing, possibly due to overfitting. In comparison, SVR managed to consistently achieve stable performance on the training, validation, and test set, implying that simpler models may be better suited when data

is comparatively small. Surprisingly poor were the weather variables, which played a poor role. In contrast to the previous investigations, the feature importance analysis indicated that the temporal features, including Year, Month, and seasonal encodings, were far more significant predictors of the monthly peak demand than weather conditions.

These variations between the assumed values and the actual ones have brought out two important things to note. First, methodological appropriateness depends on the scale and structure of data sets: less complicated models might be more effective than the advanced ones in case of few observations and inconsistency in peaks. Second, the impact of weather could dissipate on medium, aggregated, monthly scales, and, instead, cyclic patterns in temporal behaviour patterns will become predominant on the station side.

5.3 Predictability of Peak Load

The main questions were: would one be able to reliably predict monthly peak demand on EV charging stations based on historical consumption and other additional characteristics? The models obtained test R squares of 0.49 to 0.62 and SVR yielded the most reliable generalisation. Although not a significant percent increase, these values were consistently above baseline averages, and showed that peak load can at least be partially) predicted at the station level.

This result agrees with the conclusion to reject the null hypothesis (H_0) of peak load predictability. It also supports previous research that shows that historical charging behaviour is predictive of future demand (Zhang et al., 2023). Nevertheless, this level of accuracy was weaker compared to those of the short-term forecasting findings, wherein hourly and daily predictions attain R^2 scores of 0.8 or higher in accuracy (Zhu et al., 2019; Sun et al., 2016).

The key reason for this difference lies in the modelling horizon. By working with monthly aggregates, this study necessarily reduced the variability available to the models. Peaks are

often defined by short-lived events that are obscured when data is aggregated to higher levels of temporal granularity. As such, the models performed better at identifying broad seasonal trends and growth patterns than at capturing the magnitude of individual monthly peaks.

5.4 Contribution of Features: Temporal and Weather Variables

Weather variables were believed to have a powerful impact on peak demand and this expectation was a result of this study. Earlier studies on the energy demand forecasting have reported consistently that temperature, precipitation, and humidity are also major predictors in the forecast (Islam et al., 2023; Xiong et al., 2015). The rationale was that extreme weather changes driver and charging behaviour, wet weather decreases the frequency of the travel and humidity affects the comfortability and battery.

In practice, this study's feature importance analysis suggested otherwise. Temporal variables especially Year, reflecting long-term adoption growth, reflecting cyclical usage patterns emerged as more influential than any individual weather variable. While temperature showed some non-linear correlation with demand, overall, weather factors played a weaker role than anticipated.

The finding disputes conclusions made based on the general literature and indicates that the correlation between weather and EV charging demand is strongly subject to the level of representation in time. Short-term (daily, hourly) timelines have more measurable weather effects as immediate behavioural responses are possible (Zhu et al., 2019). At the monthly level, however, the effects are smoothed out by aggregation and bigger structural factors are in the lead. As an example, EV adoption level, charging station availability, and commuting habits are more likely to add to the monthly peak tendencies than the variation of rainfall or wind speed. Thus, Hypothesis 2 was partly confirmed. Temporal and seasonal variables were also deemed critical but weather-related were not important as suggested. This is a valuable addition

to the literature: it acts as a warning that one should not expect the short-term findings to be applied to the medium-term station-level forecasts directly.

5.5 Comparative model performance

Results of the comparison of models were not in line with the original hypothesis that higher model would have the best performance. XGBoost produced very high training performance (R^2 0.99) but low-test performance, suggesting over fitting. Neural Networks performed again the poorest, with a test R^2 of about 0.49. Random Forest did slightly better whereas generalisation was strong in SVR with test R^2 (0.60).

This result shows that technological advancement of algorithms is not a promising predictive feature. The dataset presented in this work was limited in terms of size because data was aggregated at monthly level and it targets station level information. In such environments, less complex algorithms with high regularisation capabilities, e.g., SVR, are more appropriate in the bias/variance trade-off. Such results confirm the limitations of deep learning approaches in the industrial environment discussed in previous studies regarding scenarios with modest volumes of data and where explainability is important (Wu et al., 2017).

The rejection of Hypothesis 3 highlights the importance of selecting models based on qualities of the dataset. Instead of falling back to deep learning or boosting, one should ask whether simpler methods can yield more reliable results in applications where available data is sparse or summarized information.

5.6 Practical Usefulness of Forecasts

Despite the moderate predictive performance, the forecasts generated in this study remain practically useful for grid operators. Seasonal trends captured by the models reveal predictable surges in summer, which can inform reserve allocation strategies. Similarly, station-level

disaggregation highlights differences in utilisation across locations, offering guidance for targeted infrastructure upgrades.

Long-term planning can benefit from the projections' directional insights, even when dramatic peaks were not always accurately caught. According to Huang et al. (2020) and Mahmoud et al. (2015), even forecasts with a moderate level of accuracy can be useful if they are placed into frameworks for making decisions. Therefore, the results validate Hypothesis 4 even though monthly peak demand estimates are not always accurate, they can nevertheless offer useful information for planning.

5.7 Interpretation of Findings

These findings reveal some more general lessons. First, the findings highlight the difficulty of monthly forecasting at the station level. High predicted accuracy is hard to obtain for models because of high volatility and little observations. Second, they demonstrate that, in contrast to a large portion of the research on short-term forecasting, temporal determinants dominate weather effects at the monthly level. This implies that distinct modelling approaches are needed for various time spans. Third, the results highlight the significance of model-data fit by showing that when data is scarce, simpler models can perform better than more complex ones. The gaps in the literature are filled by these additions. This study demonstrates the benefits of using machine learning techniques, especially interpretable ones like SVR and Random Forest, for medium-term, station-level forecasting, since the majority of previous work has concentrated on short-term horizons and complex models. The fact that the predicted predominance of weather effects was not shown here emphasises even more how crucial it is to question presumptions regarding feature relevance.

Chapter 6 Conclusion & Future Work

6.1 Conclusion

This study has shown that it is possible to estimate monthly peak electricity demand at the station level with a moderate degree of accuracy; nevertheless, there are still issues with catching extreme peaks. The predictive performance of the investigated models varied, with neural networks exhibiting an R^2 of 0.49 and support vector regression (SVR) exhibiting an R^2 of roughly 0.60. SVR demonstrated consistent generalisation across training, validation, and test sets. These results highlight the efficacy of simpler models like SVR over more complicated techniques like XGBoost and Neural Networks, which were prone to overfitting, and show that model complexity does not always translate into better performance, especially when datasets are restricted. This study's feature importance analysis, which showed that temporal variables like year and seasonal encodings were more important in forming monthly peaks than weather-related aspects, is another contribution. Instead, it suggests that aggregated horizons smooth short-term climatic influences, leaving structural and behavioural drivers like adoption growth and station utilisation as the main determinants. This is different from existing literature, which often emphasises weather as a dominant predictor. The projections were practically helpful in identifying seasonal surges and locational inequalities that can help grid operators with reserve allocation and infrastructure development, despite their limited predictive accuracy. The loss of detail due to monthly aggregation, the lack of socioeconomic and traffic-related factors, the small sample size that limits complicated models, and the restricted regional reach are some of the drawbacks, despite the fact that significant contributions have been made. Within these parameters, the study highlights how crucial it is to match modelling techniques with dataset properties and points up potential avenues for improving data robustness, richness, and application in the future.

6.2 Future work

In order to overcome the constraints noted in this work, future research should focus on growing the dataset, improving modelling techniques, and extending the range of applications. Together with traffic flow data, socioeconomic characteristics, commute intensity, and EV adoption rates could offer more explanatory power and aid in identifying behavioural drivers that were not included in this research. Additionally, future research could more successfully capture both short-term variations and long-term demand cycles by implementing multi-resolution modelling frameworks that blend short-term data (hourly or daily) with medium-term monthly aggregates. The development of methods should also focus on improving interpretability and resilience. Although they performed poorly in this study, neural networks and boosting algorithms may perform better in the future when bigger and more varied datasets are made accessible. Overfitting concerns may also be decreased by complementary strategies that combine machine learning and statistical baselines. Last but not least, cross-regional research that contrasts outcomes in various climatic, infrastructure, and adoption situations would improve the findings' generalisability and offer insightful information about how scalable forecasting models are. Future research can enhance EV peak load forecasting's predictive ability, transparency, and usefulness by tackling these issues, providing grid operators with more dependable instruments for long-term planning and the distribution of energy resources.

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